6. Normal subgroups, quotients, and extensions

A homomorphism can send different group elements to the same image. The elements sent to the identity form its kernel. Each coset of the kernel consists of all the elements with one particular image. This chapter shows that these cosets form a group isomorphic to the image. We then ask how a group can be constructed from a given normal subgroup and quotient. Rotations and translations, glide reflections, and operators carrying phases illustrate these constructions.

6.1. Normal subgroups

Reading. [Moore, §7.3, pp. 80–92].

Conjugation carries a subgroup H to another subgroup gHg^{-1}. Some subgroups are carried onto themselves by every element of G.

normal subgroup
正規子群

Definition 6.1 (Normal subgroup)**.** A subgroup $N \subseteq G$ is *normal* in G, written $N \triangleleft G$, if

$$gNg^{-1} = N \qquad \forall g \in G.$$

This is equality of sets. It permits conjugation to permute the elements of N; it does not require $gng^{-1} = n$ for each $n \in N$. That stronger condition says $N \subset Z(G)$.

Proposition 6.2. *For a subgroup $N \subset G$, the following conditions are equivalent.*

(i) $N \triangleleft G$;
(ii) $gN = Ng \quad \forall g \in G$;
(iii) N *is a union of conjugacy classes of* G.

Proof. Multiplying $gNg^{-1} = N$ on the right by g gives (ii), and the same step reversed gives (i). Condition (i) implies that the whole conjugacy class of each $n \in N$ lies in N, giving (iii). Conversely, (iii) gives $gNg^{-1} \subset N$ for every g. Applying this inclusion to g^{-1} and conjugating by g gives the reverse inclusion. □

Proposition 6.3 (Kernels are normal)**.** *If $\varphi : G \to G'$ is a homomorphism, then* $\ker \varphi \triangleleft G$.

Proof. The kernel is a subgroup by §3.2. For $k \in \ker\varphi$,

$$\varphi(gkg^{-1}) = \varphi(g)e'\varphi(g)^{-1} = e',$$

so $g(\ker\varphi)g^{-1} \subset \ker\varphi$. Applying the same argument to g^{-1} gives equality. □

Example 6.4. The subgroups $\{e\}$ and G are always normal in G. Every subgroup of an Abelian group is normal, since $ghg^{-1} = h$. The centre $Z(G)$ is normal even when G is not Abelian.

The sign homomorphism. Recall from Chapter 5 that $\mathrm{sgn} : S_n \to \{1, -1\}$ sends even permutations to 1 and odd permutations to -1, with multiplication as the group operation in $\{1, -1\}$. Its kernel consists of the even permutations, so

$$A_n = \ker(\mathrm{sgn}) \triangleleft S_n.$$

The determinant homomorphism. For $\kappa = \mathbb{R}$ or $\mathbb{C}$, the multiplicative property $\det(AB) = \det A \, \det B$ makes

$$\det : GL(n, \kappa) \longrightarrow \kappa^*$$

a homomorphism, where κ^* is the group of nonzero scalars under multiplication. Its identity is 1, so the kernel consists of the matrices with determinant 1. Thus

$$SL(n, \kappa) = \ker(\det) \triangleleft GL(n, \kappa).$$

As seen in §2.5, the determinant takes values in $\{1, -1\}$ on $O(n)$ and in $U(1)$ on $U(n)$. These restricted homomorphisms have kernels $SO(n)$ and $SU(n)$, respectively, so

$$SO(n) \triangleleft O(n), \qquad SU(n) \triangleleft U(n).$$

Proposition 6.5. *Every subgroup of index two is normal.*

Proof. If $[G : H] = 2$ and $g \notin H$, then $gH = G \setminus H = Hg$: both left and right cosets partition G into H and its complement. For $g \in H$, both cosets equal H. Apply Proposition 6.2. □

Remark 6.6 (The role of the normalizer). Definition 4.30 now has a shorter interpretation:

$$H \triangleleft N_G(H).$$

Moreover, $N_G(H)$ is the largest subgroup of G in which H is normal. Indeed, $H \triangleleft K \subset G$ implies $kHk^{-1} = H$ for every $k \in K$, hence $K \subset N_G(H)$. In particular, H is normal in G precisely when $N_G(H) = G$.

simple group
単群

Definition 6.7 (Simple group)**.** A nontrivial group G is *simple* if

$$\forall N \triangleleft G, \qquad N = \{e\} \text{ or } N = G.$$

For a prime p, $\mathbb{Z}_p$ is simple by Lagrange's theorem. The group A_4 is not simple: its identity and three double transpositions form the normal subgroup V_4. It is a subgroup by Example 5.16, and conjugation preserves cycle type. We state without proof the further result that A_n is simple for $n \geq 5$.

Quiz 6.1. In S_3, show that $\langle (1\,2\,3) \rangle$ is normal but $\langle (1\,2) \rangle$ is not. Does being Abelian imply that a subgroup is normal in the ambient group?

6.2. Quotient groups

Chapter 4 used G/H as a *set* of left cosets. Normality is exactly what allows multiplication in G to descend to multiplication of these cosets.

quotient group
商群

Theorem 6.8 (Quotient group)**.** *If $N \triangleleft G$, the rule*

$$(gN)(hN) := (gh)N$$

is well defined and makes G/N a group. Its identity is $N = eN$, and $(gN)^{-1} = g^{-1}N$. The canonical projection

$$\pi : G \longrightarrow G/N, \qquad \pi(g) = gN,$$

is a surjective homomorphism with kernel N.

Proof. Choose other representatives $g' = gn_1$ and $h' = hn_2$, with $n_1, n_2 \in N$. Then

$$g'h' = gn_1hn_2 = gh\,(h^{-1}n_1h)n_2 \in ghN,$$

since N is normal. Thus $g'h'N = ghN$, proving well-definedness. Associativity follows from that of G; N is the identity and $g^{-1}N$ is the inverse of gN. The multiplication rule says exactly that π is a homomorphism. Every coset has a representative, so π is onto, and $\pi(g) = N$ exactly when $g \in N$. □

Every normal subgroup is therefore the kernel of some homomorphism. Conversely, if this coset multiplication makes G/H a group, its canonical projection is a homomorphism with kernel H, so H must be normal. For finite G,

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

The quotient retains the distinctions between cosets and identifies elements within each coset. Its elements are cosets, not selected elements of G. In particular, a quotient need not be isomorphic to a subgroup of G.

Example 6.9 (Integers modulo N). For a positive integer N, the subgroup $N\mathbb{Z}$ is normal in the additive group $\mathbb{Z}$. The quotient has cosets

$$N\mathbb{Z},\ 1+N\mathbb{Z},\ \dots,\ (N-1)+N\mathbb{Z},$$

and addition is $(a+N\mathbb{Z})+(b+N\mathbb{Z})=(a+b)+N\mathbb{Z}$. These are exactly the residue classes of §2.3:

$$\mathbb{Z}/N\mathbb{Z}\cong\mathbb{Z}_N.$$

This explains the quotient notation introduced there. For $N>1$ it also illustrates the preceding distinction: $\mathbb{Z}$ has no nontrivial finite subgroup, although this quotient is finite.

Example 6.10 (Quotients of the square group). Write $D_4=\langle r,s\mid r^4=s^2=1,\ srs^{-1}=r^{-1}\rangle$. Its nontrivial proper normal subgroups and quotients are

N	$\lvert N\rvert$	D_4/N
$\langle r\rangle$	4	$\mathbb{Z}_2$
$\langle r^2,s\rangle$	4	$\mathbb{Z}_2$
$\langle r^2,rs\rangle$	4	$\mathbb{Z}_2$
$\langle r^2\rangle=Z(D_4)$	2	V_4

The first three are normal because their index is two. For $N=\langle r^2,s\rangle$, the two cosets are

$$N=\{e,r^2,s,r^2s\},\qquad rN=\{r,r^3,rs,r^3s\}.$$

The coset N is the identity, and $(rN)^2=r^2N=N$. Thus

$\cdot$	N	rN
N	N	rN
rN	rN	N

This is the multiplication table of $\mathbb{Z}_2$. The cases $N=\langle r\rangle$ and $N=\langle r^2,rs\rangle$ have the same table, with nonidentity cosets sN and rN, respectively.

For $Z=Z(D_4)=\langle r^2\rangle$, the four cosets are

$$Z=\{e,r^2\},\qquad rZ=\{r,r^3\},$$
$$sZ=\{s,r^2s\},\quad rsZ=\{rs,r^3s\}.$$

Each nonidentity coset squares to Z, since $r^2\in Z$ and $s^2=(rs)^2=e$. Moreover,

$$(rZ)(sZ)=rsZ,\qquad (sZ)(rZ)=srZ=r^{-1}sZ=rsZ.$$

Using $(gZ)(hZ)=(gh)Z$ gives the full table

$\cdot$	Z	rZ	sZ	rsZ
Z	Z	rZ	sZ	rsZ
rZ	rZ	Z	rsZ	sZ
sZ	sZ	rsZ	Z	rZ
rsZ	rsZ	sZ	rZ	Z

Hence $D_4/Z(D_4)\cong\mathbb{Z}_2\times\mathbb{Z}_2\cong V_4$. The four reflection subgroups of order two are not normal: conjugating $\langle r^ks\rangle$ by r gives $\langle r^{k+2}s\rangle$.

Example 6.11 (The quaternion quotient). The group Q of Example 2.25 has centre $Z = Z(Q) = \{1, -1\}$. The quotient consists of

$$Z, \quad iZ = \{\pm i\}, \quad jZ = \{\pm j\}, \quad kZ = \{\pm k\}.$$

Because $i^2 = j^2 = k^2 = -1 \in Z$, its three nonidentity elements have order two. Further, $ij = k$ and $ji = -k$ give the same coset. Explicitly,

$$Z \mapsto (0,0), \quad iZ \mapsto (1,0), \quad jZ \mapsto (0,1), \quad kZ \mapsto (1,1)$$

respects multiplication, with addition modulo two on the right. Hence

$$Q/Z(Q) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong V_4 \cong D_2.$$

The quotient has lost the signs that distinguished ij from ji. An Abelian quotient therefore does not imply an Abelian original group.

Quiz 6.2. For $H = \langle (1\,2) \rangle \subset S_3$, the elements e and $(1\,2)$ represent the same coset H. Show that $e(1\,3)$ and $(1\,2)(1\,3)$ represent different left cosets. Explain why multiplication of cosets fails.

In the homogeneous-space example $S^2 \cong SO(3)/SO(2)$, does this notation make the sphere a quotient *group*? Consider what conjugation by a rotation does to the axis fixed by $SO(2)$.

6.3. The isomorphism theorems

Reading. [Moore, §7.4, pp. 99–101].

Theorem 6.12 (First isomorphism theorem)**.** *For a homomorphism $\varphi : G \to G'$ with kernel K, the map*

$$\overline{\varphi} : G/K \longrightarrow \operatorname{im}\varphi, \qquad gK \longmapsto \varphi(g),$$

is an isomorphism.

Proof. If $gK = hK$, then $h^{-1}g \in K$, so $\varphi(g) = \varphi(h)$; the map is well defined. Moreover,

$$\overline{\varphi}((gK)(hK)) = \varphi(gh) = \overline{\varphi}(gK)\,\overline{\varphi}(hK).$$

It is onto the image by definition. If $\overline{\varphi}(gK) = e'$, then $g \in K$ and $gK = K$, so its kernel is trivial and it is injective. □

All elements in one coset of $K = \ker\varphi$ have the same image under φ, and different cosets have different images. Thus $\overline{\varphi}$ gives a bijection between the kernel cosets in G/K and the elements of $\operatorname{im}\varphi$.

Example 6.13 (Inner automorphisms). The homomorphism $I : G \to \mathrm{Aut}(G)$ from §4.4, given by $I_g(x) = gxg^{-1}$, has kernel $Z(G)$ and image $\mathrm{Inn}(G)$. Therefore

$$G/Z(G) \cong \mathrm{Inn}(G), \qquad gZ(G) \longmapsto I_g.$$

Central elements cannot be distinguished by their conjugation action. For the quaternion group this gives $\mathrm{Inn}(Q) \cong V_4$.

Example 6.14 (Determinants and rotations). The determinant maps below are surjective, with the kernels already identified in §6.1. Thus, for $n \geq 1$,

$$\begin{aligned} GL(n,\kappa)/SL(n,\kappa) &\cong \kappa^*, \\ U(n)/SU(n) &\cong U(1), \\ O(n)/SO(n) &\cong \mathbb{Z}_2. \end{aligned}$$

Surjectivity follows by using diagonal matrices with the desired determinant. For $n \geq 2$, the sign map similarly gives $S_n/A_n \cong \mathbb{Z}_2$. Theorem 3.16 now takes the quotient form

$$SU(2)/\{\pm\mathbb{1}\} \cong SO(3).$$

This uses the matrix homomorphism proved in Chapter 3.

Two further useful forms of the theorem describe restriction to a subgroup and taking quotients in stages. Their proofs are applications of the first theorem.

Theorem 6.15 (Second isomorphism theorem)**.** *If $H \subset G$ is a subgroup and $N \triangleleft G$, then the product set $HN = \{hn : h \in H,\ n \in N\}$ is a subgroup, $N \triangleleft HN$, $H \cap N \triangleleft H$, and*

$$H/(H \cap N) \cong HN/N.$$

Proof. Let $\pi : G \to G/N$ be the quotient projection, $\pi(g) = gN$. The image $\pi(H)$ is a subgroup of G/N. An element $g \in G$ lies in $\pi^{-1}(\pi(H))$ precisely when $gN = hN$ for some $h \in H$, or equivalently $g = hn$ for some $n \in N$. Thus $\pi^{-1}(\pi(H)) = HN$. If $x, y \in HN$, then $\pi(x), \pi(y) \in \pi(H)$, so $\pi(xy^{-1}) = \pi(x)\pi(y)^{-1} \in \pi(H)$ and hence $xy^{-1} \in HN$. The subgroup test of Lemma 2.5 therefore shows that HN is a subgroup of G. Since $N \triangleleft G$, we also have $N \triangleleft HN$.

Now restrict the projection to H,

$$f = \pi|_H : H \longrightarrow G/N, \qquad f(h) = hN.$$

Its kernel is

$$\ker f = \{h \in H : hN = N\} = H \cap N,$$

so $H \cap N \triangleleft H$. Its image consists of the cosets hN with $h \in H$. Every element of HN has the form hn, and $(hn)N = hN$. Consequently, using representatives from HN gives exactly the same cosets as using representatives from H, so

$$\mathrm{im}\, f = \{hN : h \in H\} = HN/N.$$

Applying the first isomorphism theorem to f gives

$$H/(H \cap N) \cong \mathrm{im}\, f = HN/N, \qquad h(H \cap N) \longmapsto hN.$$

□

Example 6.16. For $G = S_3$, $N = A_3$, and $H = \langle (1\,2) \rangle$, we have $H \cap N = \{e\}$ and $HN = S_3$. Thus $S_3/A_3 \cong H \cong \mathbb{Z}_2$. Here the quotient projection restricts to an isomorphism $H \to S_3/A_3$.

Theorem 6.17 (Third isomorphism theorem)**.** *If $N \subset H \subset G$ with both N and H normal in G, then $H/N \triangleleft G/N$ and*

$$(G/N)/(H/N) \cong G/H.$$

Proof. Define

$$f : G/N \longrightarrow G/H, \qquad f(gN) = gH.$$

If $gN = g'N$, then $g^{-1}g' \in N \subset H$, so $gH = g'H$. Thus f is well defined. It is a homomorphism because

$$f((gN)(g'N)) = (gg')H = (gH)(g'H) = f(gN)f(g'N),$$

and it is surjective because every coset gH is the image of gN.

An element gN lies in the kernel precisely when $gH = H$, that is, when $g \in H$. Hence

$$\ker f = \{hN : h \in H\} = H/N.$$

As the kernel of a homomorphism, H/N is normal in G/N. The first isomorphism theorem (Theorem 6.12) therefore gives

$$(G/N)/(H/N) \cong G/H, \qquad (gN)(H/N) \longmapsto gH.$$

Thus taking the quotient first by N and then by H/N gives the same group as taking the quotient directly by H. □

Quiz 6.3. If $G/Z(G)$ is cyclic, prove that G is Abelian. Hint: if $gZ(G)$ generates the quotient, every element of G is $g^k z$ for some $z \in Z(G)$.

Use the third theorem with $4\mathbb{Z} \subset 2\mathbb{Z} \subset \mathbb{Z}$ to identify $\mathbb{Z}_4/\{\overline{0}, \overline{2}\}$. Explain each map rather than only comparing group orders.

Reading. [Moore, §7.4, pp. 98–103, §14, pp. 408–415, §15.1, pp. 423–430].

6.4. Exact sequences and group extensions

Example 6.18 (Forgetting the translation). Return to the spatial transformations of Example 4.6. Restricting R to $SO(3)$ gives the orientation-preserving Euclidean group, denoted $SE(3)$. Its elements act by $x \mapsto Rx + a$ and multiply as

$$\{R \mid a\}\{S \mid b\} = \{RS \mid a + Rb\}.$$

The map $\pi(\{R \mid a\}) = R$ keeps the rotation and forgets the translation. It is onto $SO(3)$, and its kernel is the subgroup of pure translations $\{\mathbb{1} \mid a\}$, isomorphic to $(\mathbb{R}^3, +)$. Hence

$$SE(3)/\mathbb{R}^3 \cong SO(3).$$

The notation $\mathbb{R}^3$ in this quotient means its embedded translation subgroup. The information can be displayed together as

$$1 \longrightarrow \mathbb{R}^3 \xrightarrow{\iota} SE(3) \xrightarrow{\pi} SO(3) \longrightarrow 1, \tag{6.1}$$

where $\iota(a) = \{\mathbb{1} \mid a\}$. Every rotation occurs, and the transformations mapped to the identity rotation are exactly the translations. Including reflections gives the analogous sequence with $E(3)$ and $O(3)$.

exact sequence
正合列

Definition 6.19 (Exact sequence)**.** A sequence of groups and homomorphisms has the form

$$\cdots \longrightarrow G_{i-1} \xrightarrow{f_{i-1}} G_i \xrightarrow{f_i} G_{i+1} \longrightarrow \cdots,$$

where each $f_i : G_i \to G_{i+1}$ is a homomorphism. It is *exact* at G_i if the image of the incoming map equals the kernel of the outgoing map,

$$\operatorname{im} f_{i-1} = \ker f_i.$$

The sequence is *exact* if this condition holds at every intermediate group. A *short exact sequence* has the form

$$1 \longrightarrow N \xrightarrow{\iota} E \xrightarrow{\pi} G \longrightarrow 1 \tag{6.2}$$

and is exact at N, E, and G. Here 1 denotes the trivial group.

The map from 1 has image $\{e_N\}$, so exactness at N means $\ker \iota = \{e_N\}$: ι is injective. The map to 1 has kernel all of G, so exactness at G means π is onto. Exactness at E is the condition $\operatorname{im} \iota = \ker \pi$. Thus the short exact sequence (6.2) gives

$$N \cong \iota(N) \triangleleft E, \qquad E/\iota(N) \cong G.$$

The quotient isomorphism follows from the first isomorphism theorem (Theorem 6.12) applied to π, since $\ker \pi = \iota(N)$ and $\operatorname{im} \pi = G$. For sequences of Abelian groups, we use additive notation and denote the trivial group by 0.

group extension
群扩张

Definition 6.20 (Group extension)**.** A short exact sequence of the form (6.2) is an *extension of G by N*. We call N its kernel group and G its quotient group. We usually identify N with its image $\iota(N) \subset E$.

The sequence describes which information is forgotten by π. The extension problem asks how that information enters multiplication in E. Specifying the abstract groups N and G alone need not determine E. For finite groups it determines $|E| = |N||G|$, but not its group law.

Example 6.21 (The same kernel and quotient). Both of the following sequences are exact:

$$0 \longrightarrow \mathbb{Z}_2 \xrightarrow{a\mapsto 2a} \mathbb{Z}_4 \xrightarrow{b\mapsto b \bmod 2} \mathbb{Z}_2 \longrightarrow 0,$$

$$0 \longrightarrow \mathbb{Z}_2 \xrightarrow{a\mapsto(a,0)} \mathbb{Z}_2 \times \mathbb{Z}_2 \xrightarrow{(a,b)\mapsto b} \mathbb{Z}_2 \longrightarrow 0.$$

The middle groups are not isomorphic: $\mathbb{Z}_4$ has an element of order four, whereas every nonidentity element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order two.

6.4.1. Split extensions and semidirect products

The first isomorphism theorem identifies E/N with G by $xN \mapsto \pi(x)$. A *lift* of $g \in G$ is an element $x \in E$ satisfying $\pi(x) = g$. All lifts of g form the *fiber* $\pi^{-1}(g)$, which is one coset of the kernel N.

A *section* is a map $s : G \to E$ that assigns one lift $s(g) \in E$ to each element $g \in G$,

$$\pi(s(g)) = g \qquad \forall g \in G.$$

Thus s is a right inverse of π, and $\pi^{-1}(g) = s(g)N$. Under the identification $E/N \cong G$, the chosen lift $s(g)$ is a representative of the coset corresponding to g.

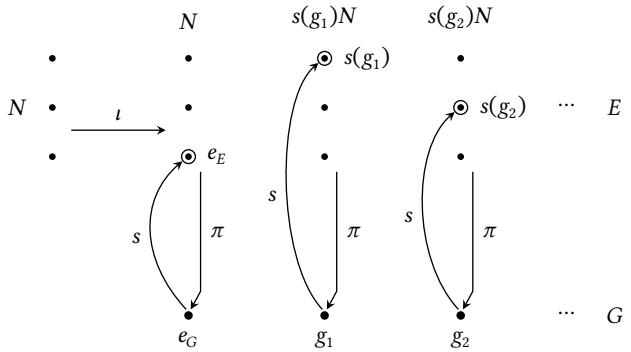
split extension
分裂扩张

Definition 6.22 (Splitting)**.** The short exact sequence (6.2) *splits* if

$$\exists \text{ homomorphism } s : G \to E \quad \text{with} \quad \pi \circ s = \mathrm{id}_G .$$

Such a homomorphic section is called a *splitting*.

The identity $\pi \circ s = \mathrm{id}_G$ makes s injective, since $s(g) = s(h)$ implies $g = \pi(s(g)) = \pi(s(h)) = h$. Its image $s(G)$ is a subgroup of E, and the restriction $\pi|_{s(G)} : s(G) \to G$ is an isomorphism.



Each column in E is a coset of N, projected to one element of G. The splitting s selects the circled lift in each coset.

In the $\mathbb{Z}_4$ sequence, the projection is $\pi(x) = x \bmod 2$. The nonzero element $1 \in \mathbb{Z}_2$ has two preimages, 1 and 3 in $\mathbb{Z}_4$, each of order four. A splitting must choose one of these as $s(1)$. Since $1 + 1 = 0$ in $\mathbb{Z}_2$, a homomorphism would require

$$s(1) + s(1) = s(1 + 1) = s(0) = 0.$$

But $1 + 1 = 2$ and $3 + 3 = 2$ in $\mathbb{Z}_4$. Thus neither choice preserves addition, and the extension does not split.

For the $\mathbb{Z}_2 \times \mathbb{Z}_2$ sequence, the projection is $\pi(a, b) = b$. Choose $s(b) = (0, b)$. Then

$$\pi(s(b)) = b,$$
$$s(b + b') = (0, b + b') = (0, b) + (0, b') = s(b) + s(b').$$

Thus s is a homomorphism with $\pi \circ s = \mathrm{id}_{\mathbb{Z}_2}$, so this extension splits. Here $s(1) = (0, 1)$ has order two. Another splitting is $s_1(b) = (b, b)$, which also preserves addition.

Return now to the Euclidean sequence (6.1), with $N = (\mathbb{R}^3, +)$, $E = SE(3)$, and $G = SO(3)$. Write

$$T_a = \{\mathbb{1} \mid a\}, \qquad s(R) = \{R \mid 0\}.$$

Here $\pi(s(R)) = R$ and $s(R)s(S) = s(RS)$, so s is a splitting. Every element of $SE(3)$ has the factorization

$$\{R \mid a\} = T_a s(R).$$

The projection determines R. After cancelling $s(R)$, the remaining translation determines a, so the factorization is unique.

To multiply two such factorizations, we must move a translation past a rotation. Conjugating T_b by $s(R)$ gives

$$\left[s(R)T_b s(R)^{-1}\right](x) = R(R^{-1}x + b) = x + Rb.$$

Thus $s(R)T_b s(R)^{-1} = T_{Rb}$, or equivalently $s(R)T_b = T_{Rb}s(R)$. The translation vector changes from b to Rb when we reverse the order of the two operations. Consequently,

$$\begin{aligned}\big(T_a s(R)\big)\big(T_b s(S)\big) &= T_a T_{Rb} s(R)s(S) \\ &= T_{a+Rb} s(RS).\end{aligned}$$

In pair notation this is $(a, R)(b, S) = (a + Rb, RS)$. The following theorem gives the same factorization and multiplication for a general split extension.

Theorem 6.23 (Structure of a split extension)**.** *Let $s : G \to E$ be a splitting of* (6.2)*. Identify N with its image $\iota(N) \triangleleft E$. Conjugation by $s(g)$ defines*

$$\alpha_g : N \longrightarrow N, \qquad \alpha_g(n) = s(g)ns(g)^{-1}.$$

Then $\alpha : G \to \mathrm{Aut}(N)$, $g \mapsto \alpha_g$, is a homomorphism. Every $x \in E$ has a unique expression $x = \iota(n)s(g)$, with $n \in N$ and $g \in G$. The product is taken in E and is abbreviated $ns(g)$ after identifying N with $\iota(N)$. Thus the map

$$\Phi : N \times G \longrightarrow E, \qquad (n, g) \longmapsto \iota(n)s(g),$$

is a bijection of sets. Under this bijection, multiplication in E becomes

$$(n, g)(m, h) = \big(n\alpha_g(m), gh\big). \tag{6.3}$$

With this multiplication, $N \times G$ is a group and Φ is an isomorphism. The factorization $(n, g) = (n, e_G)(e_N, g)$ corresponds to $\iota(n)s(g)$ in E.

Proof. Conjugation by $s(g)$ maps the normal subgroup N onto itself, with inverse given by conjugation by $s(g)^{-1}$. Thus $\alpha_g \in \mathrm{Aut}(N)$. Since $s(gh) = s(g)s(h)$,

$$\begin{aligned} \alpha_{gh}(n) &= s(g)s(h)ns(h)^{-1}s(g)^{-1} \\ &= \alpha_g\big(\alpha_h(n)\big). \end{aligned}$$

Hence $\alpha_{gh} = \alpha_g \circ \alpha_h$, proving that $\alpha : G \to \mathrm{Aut}(N)$ is a homomorphism.

For $x \in E$, put $g = \pi(x)$. Then $xs(g)^{-1} \in \ker \pi = \iota(N)$, giving $x = \iota(n)s(g)$. If $\iota(n)s(g) = \iota(m)s(h)$, applying π gives $g = h$, then injectivity of ι gives $n = m$. Thus Φ is a bijection. Finally, using $s(g)\iota(m)s(g)^{-1} = \iota(\alpha_g(m))$,

$$\begin{aligned} \Phi(n, g)\Phi(m, h) &= \iota(n)s(g)\iota(m)s(h) \\ &= \iota\big(n\alpha_g(m)\big)s(gh) \\ &= \Phi\big(n\alpha_g(m), gh\big). \end{aligned}$$

Hence multiplication in E gives precisely the pair multiplication in (6.3). Transporting the group law through the bijection Φ makes $N \times G$ a group isomorphic to E. □

semidirect product
半直积

Definition 6.24 (Semidirect product)**.** Let $\alpha : G \to \mathrm{Aut}(N)$ be a homomorphism, with $\alpha_g = \alpha(g)$. The *semidirect product* $N \rtimes_\alpha G$ is the set $N \times G$ of ordered pairs (n, g), with $n \in N$ and $g \in G$, equipped with the multiplication (6.3).

The identity is (e_N, e_G) and

$$(n, g)^{-1} = \big(\alpha_{g^{-1}}(n^{-1}), g^{-1}\big).$$

Since $\alpha_{gh} = \alpha_g \circ \alpha_h$, either bracketing of $(n, g)(m, h)(\ell, k)$ gives

$$\big(n\alpha_g(m)\alpha_{gh}(\ell), ghk\big),$$

which proves associativity. The maps $\iota(n) = (n, e_G)$ and $\pi(n, g) = g$ give the short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} N \rtimes_\alpha G \xrightarrow{\pi} G \longrightarrow 1.$$

It splits with $s(g) = (e_N, g)$. Thus every split extension is isomorphic to a semidirect product, and every semidirect product gives a split extension. If every α_g is the identity map of N, this is the direct product $N \times G$.

Example 6.25 (Euclidean and square-lattice groups). The Euclidean calculation above now gives

$$SE(3) \cong \mathbb{R}^3 \rtimes_\alpha SO(3), \qquad \alpha_R(b) = Rb.$$

Different choices of origin give different splittings. A rotation about $v \in \mathbb{R}^3$ acts as $x \mapsto v + R(x - v)$, giving

$$s_v(R) = \{R \mid (\mathbb{1} - R)v\}, \qquad \pi(s_v(R)) = R.$$

The translation part of $s_v(R)s_v(S)$ is

$$(\mathbb{1} - R)v + R(\mathbb{1} - S)v = (\mathbb{1} - RS)v,$$

so $s_v(R)s_v(S) = s_v(RS)$. Each origin therefore gives a splitting of the same extension.

The same conjugation calculation on the square lattice recovers Example 4.23:

$$p4mm \cong \mathbb{Z}^2 \rtimes D_4.$$

Here every point-group operation has a representative with zero translation, and these representatives form a subgroup.

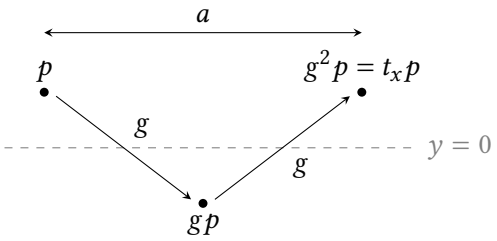
Reading. [Moore, §15.6.1, pp. 519–522].

Example 6.26 (A nonsplit spatial extension). Take $a, b > 0$ and let Γ be the group generated on the plane by

$$t_x(x, y) = (x + a, y), \quad t_y(x, y) = (x, y + b), \quad g(x, y) = (x + a/2, -y).$$

The operation g is a glide reflection. Direct composition gives

$$g^2 = t_x, \qquad g t_x g^{-1} = t_x, \qquad g t_y g^{-1} = t_y^{-1}.$$



Consequently every element has the form $t_x^m t_y^n$ or $t_x^m t_y^n g$, with $m, n \in \mathbb{Z}$. Write a spatial transformation as $\mathrm{r} \mapsto A\mathrm{r} + \mathrm{t}$, where A is a rotation or reflection matrix and t is a translation vector. The projection π keeps A and discards t. Here

$$\pi(t_x^m t_y^n) = \mathbb{1}, \qquad \pi(t_x^m t_y^n g) = M, \qquad M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its kernel is the full translation subgroup $T = \langle t_x, t_y \rangle \cong \mathbb{Z}^2$, giving

$$1 \longrightarrow T \longrightarrow \Gamma \xrightarrow{\pi} \{\mathbb{1}, M\} \longrightarrow 1.$$

The quotient reflection has order two. However, every representative of it has square

$$\begin{aligned}(t_x^m t_y^n g)^2 &= t_x^m t_y^n \big(g t_x^m t_y^n g^{-1}\big) g^2 \\ &= t_x^m t_y^n t_x^m t_y^{-n} t_x = t_x^{2m+1} \neq e.\end{aligned}$$

No representative has order two, so no homomorphism s can split this sequence. A point-group reflection need not be an actual reflection in this group: here it must come with half a translation.

A space group is *symmorphic* if and only if its point-group projection admits a homomorphic choice of representatives for the whole point group, so the extension by its full translation lattice splits. If no such choice exists, it is *nonsymmorphic*, as in this glide example. The translation and point groups alone therefore do not determine the space group.

6.4.2. Central extensions

central extension
中心扩张

Definition 6.27 (Central extension)**.** An extension

$$1 \longrightarrow A \longrightarrow E \xrightarrow{\pi} G \longrightarrow 1$$

is *central* if its kernel subgroup A lies in $Z(E)$.

The kernel group A is then Abelian, but E need not be. For example, Q is a central extension of V_4 by $\{\pm 1\}$, as Example 6.11 shows. Likewise, $SU(2)$ is a central extension of $SO(3)$ by $\{\pm \mathbb{1}\} \cong \mathbb{Z}_2$, using the homomorphism $R : SU(2) \to SO(3)$ of Theorem 3.16. Its kernel consists of scalar matrices, which commute with every element of $SU(2)$.

Since $A \subset Z(E)$, conjugation by a lift leaves each $a \in A$ unchanged. If the extension splits, the action α in Theorem 6.23 is therefore trivial, so $E \cong A \times G$. Centrality alone does not imply splitting, as the $\mathbb{Z}_4$ extension in Example 6.21 shows.

Quiz 6.4. Write $D_n \cong \mathbb{Z}_n \rtimes \mathbb{Z}_2$ explicitly: which automorphism of $\mathbb{Z}_n$ does the nonidentity element of $\mathbb{Z}_2$ induce? For which $n \geq 2$ does this become a direct product?

Optional. Verify $\varphi I_g \varphi^{-1} = I_{\varphi(g)}$ for $\varphi \in \mathrm{Aut}(G)$. Deduce $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$, define $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$, and check exactness of

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{I} \mathrm{Aut}(G) \longrightarrow \mathrm{Out}(G) \longrightarrow 1.$$

6.5. The Weyl–Heisenberg group

Central extensions, introduced in §6.4, occur naturally in quantum mechanics. Position and momentum displacement operators give an extension of the additive displacement group $(\mathbb{R}^2, +)$ by the scalar phase group $U(1)$. This is the Weyl–Heisenberg group. Clock and shift matrices give its finite counterpart.

Reading. [Moore, §7.4, pp. 102–103, §15.5.1, pp. 479–481].

6.5.1. The continuous Weyl–Heisenberg group

For a particle on the line, the position-space wavefunction $\psi(x)$ is a complex function with finite squared norm. The inner product and normalization condition are

$$\langle \chi, \psi \rangle = \int_{\mathbb{R}} \overline{\chi(x)} \psi(x)\, dx, \qquad \|\psi\|^2 = \langle \psi, \psi \rangle = 1.$$

The probability density for position is $|\psi(x)|^2$.

The position and momentum operators act on smooth, rapidly decaying wavefunctions by

$$(\hat{x}\psi)(x) = x\psi(x), \qquad (\hat{p}\psi)(x) = -\mathrm{i}\hbar \frac{d\psi}{dx}.$$

Here $\hbar$ is Planck's constant divided by 2π. For two operators, their *commutator* is $[A, B] = AB - BA$. The product rule gives

$$([\hat{x}, \hat{p}]\psi)(x) = -\mathrm{i}\hbar x \frac{d\psi}{dx}(x) + \mathrm{i}\hbar \frac{d}{dx}\big(x\psi(x)\big) = \mathrm{i}\hbar\psi(x).$$

Thus the *canonical commutation relation* is

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar \mathbb{1}, \tag{6.4}$$

where $\mathbb{1}$ is the identity operator. A plane wave with wave number k and momentum $p_0 = \hbar k$ is

$$\psi_{p_0}(x) = e^{\mathrm{i}p_0 x/\hbar}, \qquad \hat{p}\psi_{p_0} = p_0 \psi_{p_0}.$$

These plane waves are not normalizable on the whole line; normalizable wavefunctions are formed by superposing them. The momentum-space wavefunction $\phi(p)$ is the Fourier transform of the position-space wavefunction $\psi(x)$,

$$\begin{aligned} \phi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} \psi(x) e^{-\mathrm{i}px/\hbar}\, dx, \\ \psi(x) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} \phi(p) e^{\mathrm{i}px/\hbar}\, dp. \end{aligned} \tag{6.5}$$

This normalization gives $\int_{\mathbb{R}} |\phi(p)|^2\, dp = \int_{\mathbb{R}} |\psi(x)|^2\, dx$; $|\phi(p)|^2$ is the probability density for momentum. In this description, the same observables act as

$$(\hat{p}\phi)(p) = p\phi(p), \qquad (\hat{x}\phi)(p) = \mathrm{i}\hbar \frac{d\phi}{dp}.$$

These formulas follow by integration by parts and differentiation of (6.5), respectively.

We now translate position by a or momentum by b. Recall the induced action of Definition 4.36, $(g \cdot f)(u) = f(g^{-1} \cdot u)$. The inverse translations subtract a and b, so the induced operators are

$$(T_a\psi)(x) = \psi(x-a), \qquad (M_b\phi)(p) = \phi(p-b). \tag{6.6}$$

For example, a packet centred at x_0 is moved by T_a to a packet centred at $x_0 + a$; M_b shifts the momentum distribution by b.

To compose the two operators, we need their formulas in the same coordinate space. The inverse Fourier transform gives the position-space form of M_b. Substituting $q = p - b$,

$$\begin{aligned}(M_b\psi)(x) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} \phi(p-b) e^{\mathrm{i}px/\hbar}\, dp \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} \phi(q) e^{\mathrm{i}(q+b)x/\hbar}\, dq = e^{\mathrm{i}bx/\hbar}\psi(x).\end{aligned}$$

Similarly, substituting $y = x - a$ in the Fourier transform gives

$$(T_a\phi)(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} \psi(x-a) e^{-\mathrm{i}px/\hbar}\, dx = e^{-\mathrm{i}ap/\hbar}\phi(p).$$

Each family obeys an additive group law,

$$T_aT_{a'} = T_{a+a'}, \qquad M_bM_{b'} = M_{b+b'}, \qquad T_0 = M_0 = \mathbb{1}.$$

Thus $T_a^{-1} = T_{-a}$ and $M_b^{-1} = M_{-b}$. An invertible linear operator is *unitary* if it preserves the inner product, and hence normalization. A change of variable preserves the inner product for T_a, and the phase in M_b cancels its complex conjugate. Both operators are therefore unitary.

For the plane wave $\psi_{p_0}(x) = e^{\mathrm{i}p_0x/\hbar}$, these displacements take a particularly simple form,

$$\begin{aligned}(T_a\psi_{p_0})(x) &= \psi_{p_0}(x-a) = e^{\mathrm{i}p_0(x-a)/\hbar} = e^{-\mathrm{i}ap_0/\hbar}\psi_{p_0}(x), \\ (M_b\psi_{p_0})(x) &= e^{\mathrm{i}(p_0+b)x/\hbar} = \psi_{p_0+b}(x).\end{aligned}$$

Thus M_b increases the momentum label by b. Translating a plane wave by a gives a scalar phase that depends on its momentum.

These formulas identify the generators. Since $\hat{p}$ has eigenvalue p_0 on ψ_{p_0}, its exponential $e^{-\mathrm{i}a\hat{p}/\hbar}$ multiplies that wave by $e^{-\mathrm{i}ap_0/\hbar}$. It therefore agrees with T_a on every plane-wave component and on their Fourier superpositions. Likewise, $\hat{x}$ multiplies by x, so $e^{\mathrm{i}b\hat{x}/\hbar}$ multiplies by $e^{\mathrm{i}bx/\hbar}$, exactly as M_b does. Hence

$$T_a = e^{-\mathrm{i}a\hat{p}/\hbar}, \qquad M_b = e^{\mathrm{i}b\hat{x}/\hbar}. \tag{6.7}$$

Their action on the observables follows directly from the induced translations. For position,

$$\begin{aligned}(T_a^{-1}\hat{x}T_a\psi)(x) &= (\hat{x}T_a\psi)(x+a) \\ &= (x+a)(T_a\psi)(x+a) = (x+a)\psi(x).\end{aligned}$$

The first step evaluates the *whole function* $\hat{x}T_a\psi$ at $x+a$, so its position factor is also $x+a$. For momentum, use $\phi(p)$, where $\hat{p}$ acts by multiplication by p. The same calculation gives

$$\begin{aligned}(M_b^{-1}\hat{p}M_b\phi)(p) &= (\hat{p}M_b\phi)(p+b)\\ &= (p+b)(M_b\phi)(p+b) = (p+b)\phi(p).\end{aligned}$$

Also, T_a commutes with $\hat{p}$ because both act by multiplication in momentum space; M_b commutes with $\hat{x}$ for the same reason in position space. Thus

$$\begin{aligned}T_a^{-1}\hat{x}T_a &= \hat{x}+a\mathbb{1}, \qquad T_a^{-1}\hat{p}T_a = \hat{p},\\ M_b^{-1}\hat{p}M_b &= \hat{p}+b\mathbb{1}, \quad M_b^{-1}\hat{x}M_b = \hat{x}.\end{aligned}$$

The position conjugation formula now determines how T_a conjugates M_b. The same induced-action calculation replaces the argument of any multiplication operator $f(\hat{x})$ by $x+a$. Taking $f(x)=e^{\mathrm{i}bx/\hbar}$ gives

$$\begin{aligned}T_a^{-1}M_bT_a &= T_a^{-1}e^{\mathrm{i}b\hat{x}/\hbar}T_a = e^{\mathrm{i}b(T_a^{-1}\hat{x}T_a)/\hbar}\\ &= e^{\mathrm{i}b(\hat{x}+a\mathbb{1})/\hbar} = e^{\mathrm{i}ab/\hbar}M_b.\end{aligned}$$

Weyl relation
外尔关系

The last step separates the scalar phase from $e^{\mathrm{i}b\hat{x}/\hbar}$. Multiplying on the left by T_a gives the *Weyl relation*,

$$M_bT_a = e^{\mathrm{i}ab/\hbar}T_aM_b. \tag{6.8}$$

The scalar factor records the dependence on the order of the two displacements. Including all such phases gives a group.

Definition 6.28 (Weyl–Heisenberg group)**.** The *Weyl–Heisenberg group* for one position–momentum pair is the operator group

$$\mathscr{W} = \{zT_aM_b \,:\, z\in U(1),\ a,b\in\mathbb{R}\}.$$

Its multiplication is composition of operators. With the central parameter taken as a phase in $U(1)$, this version is also called the *reduced Heisenberg group*.

For the multiplication law, use the Weyl relation to move M_b past $T_{a'}$,

$$\begin{aligned}(zT_aM_b)(wT_{a'}M_{b'}) &= zwT_a(M_bT_{a'})M_{b'}\\ &= zwe^{\mathrm{i}ba'/\hbar}T_{a+a'}M_{b+b'}.\end{aligned} \tag{6.9}$$

The identity is $T_0M_0=\mathbb{1}$, and

$$(zT_aM_b)^{-1} = z^{-1}e^{\mathrm{i}ab/\hbar}T_{-a}M_{-b},$$

so these operators form a group. The displacement labels a,b and the phase z are uniquely determined by the action on wavefunctions. A nonzero wavefunction with bounded support determines a from the shift of its support. After cancelling T_a, equality of the multiplication factors $ze^{\mathrm{i}bx/\hbar}$ for every x determines b and z.

Proposition 6.29 (The continuous central extension)**.** *The centre of $\mathscr{W}$ is the scalar phase subgroup $Z(\mathscr{W}) = \{z\mathbb{1} : z \in U(1)\}$. The projection $\pi(zT_aM_b) = (a, b)$ gives a nonsplit central extension*

$$1 \longrightarrow U(1) \xrightarrow{z\mapsto z\mathbb{1}} \mathscr{W} \xrightarrow{\pi} (\mathbb{R}^2, +) \longrightarrow 1. \tag{6.10}$$

Consequently $\mathscr{W}/U(1) \cong (\mathbb{R}^2, +)$.

Proof. The product (6.9) adds the displacement labels, so π is a surjective homomorphism. Its kernel consists exactly of the scalar phases. For $U = zT_aM_b$, the Weyl relation gives

$$UT_c = e^{\mathrm{i}bc/\hbar}T_cU, \qquad M_dU = e^{\mathrm{i}ad/\hbar}UM_d.$$

Commuting with every T_c forces $b = 0$, and commuting with every M_d forces $a = 0$. Thus the kernel is the whole centre. The quotient isomorphism follows from the first isomorphism theorem.

For a splitting, the commuting displacements $(a, 0)$ and $(0, b)$ would need commuting lifts. Every pair of lifts has the form zT_a and wM_b, but

$$(wM_b)(zT_a) = e^{\mathrm{i}ab/\hbar}(zT_a)(wM_b).$$

Choose a, b with $e^{\mathrm{i}ab/\hbar} \neq 1$. No choices of z, w make these lifts commute, so no homomorphic section exists. □

In $\mathscr{W}/U(1)$, a coset consists of all operators zT_aM_b with the same pair (a, b). The two products M_bT_a and T_aM_b lie in that same coset, because their ratio is the scalar phase $e^{\mathrm{i}ab/\hbar}\mathbb{1}$ in the kernel. Consequently their quotient elements commute.

6.5.2. The finite Weyl–Heisenberg group

Reading. [Moore, §11.17.2, pp. 314–317].

For a finite counterpart, constrain the particle to $N \geq 2$ equally spaced sites $x_j = jL/N$ on a ring of circumference L. Identify these sites with the points

$$z_j = e^{2\pi\mathrm{i}x_j/L} = \omega^j, \qquad \omega = e^{2\pi\mathrm{i}/N}, \qquad \mu_N = \{1, \omega, \dots, \omega^{N-1}\}.$$

A wavefunction $\psi : \mu_N \to \mathbb{C}$ assigns a complex amplitude $\psi_j = \psi(\omega^j)$ to each site. It is therefore an N-component vector,

$$\psi \longleftrightarrow \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \in \mathbb{C}^N.$$

We use the usual inner product of complex vectors,

$$\langle \chi, \psi \rangle = \sum_{j=0}^{N-1} \overline{\chi_j}\psi_j = \sum_{j=0}^{N-1} \overline{\chi(\omega^j)}\psi(\omega^j).$$

For a normalized wavefunction, $|\psi_j|^2$ is the probability of finding the particle at site j, so

$$\langle \psi, \psi \rangle = \sum_{j=0}^{N-1} |\psi_j|^2 = 1.$$

The rotation $z \mapsto \omega z$ moves the particle forward by one site. Let P be its induced action on wavefunctions, and let Q multiply by the position-dependent phase z,

$$(P\psi)(z) = \psi(\omega^{-1} z), \qquad (Q\psi)(z) = z\psi(z), \qquad z \in \mu_N. \tag{6.11}$$

Thus P is the finite counterpart of T_a with $a = L/N$. For $b = 2\pi\hbar/L$, the phase $e^{\mathrm{i}bx_j/\hbar}$ equals z_j, so Q is the counterpart of M_b. Both operators are unitary. The operator P permutes the sites, while Q multiplies each amplitude by a number of modulus one. Both have order N.

To find their multiplication relation, use the same induced-action calculation as in the continuous case,

$$\begin{aligned} (P^{-1}QP\psi)(z) &= (QP\psi)(\omega z) \\ &= \omega z (P\psi)(\omega z) \\ &= \omega z \psi(z) = \omega (Q\psi)(z). \end{aligned}$$

Hence $P^{-1}QP = \omega Q$, giving the finite Weyl relation

$$QP = \omega PQ, \qquad QPQ^{-1}P^{-1} = \omega \mathbb{1}. \tag{6.12}$$

Thus any group containing P and Q also contains the scalar phase $\omega\mathbb{1}$.

The states localized at individual sites form an orthonormal basis $e_0, \dots, e_{N-1}$, defined by

$$e_j(\omega^k) = \delta_{jk}, \qquad \psi = \sum_{j=0}^{N-1} \psi(\omega^j) e_j.$$

The operator formulas give

$$Pe_j = e_{j+1}, \qquad Qe_j = \omega^j e_j,$$

where all labels are taken modulo N. These are the *shift* and *clock* matrices in the site basis. For $N = 3$,

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}.$$

Heisenberg group
海森堡群

Definition 6.30 (Finite Weyl–Heisenberg group)**.** The *finite Weyl–Heisenberg group*, also called the *finite Heisenberg group*, is the group of unitary operators

$$\mathrm{Heis}_N = \langle P, Q, \omega\mathbb{1} \rangle \subset U(N).$$

The $N = 3$ construction appears in Sylvester's 1882 work on *nonions* [Sylvester]. Weyl [Weyl27, §8] used finite clock and shift operators in his 1927 treatment of quantum mechanics, before passing to continuous groups.

Proposition 6.31. *Every element of* Heis_N *has a unique form* $\omega^a P^b Q^c$*, with* $a, b, c \in \mathbb{Z}_N$*. Consequently* $|\mathrm{Heis}_N| = N^3$*. Writing an element as* (a, b, c)*, the multiplication is*

$$(a, b, c)(a', b', c') = (a + a' + cb',\ b + b',\ c + c'), \tag{6.13}$$

with all entries reduced modulo N*. The term* cb' *is the finite counterpart of the phase* $e^{iba'/\hbar}$ *in* (6.9)*.*

Proof. The relation $Q^c P^{b'} = \omega^{cb'} P^{b'} Q^c$ moves every P to the left of every Q, proving existence and the multiplication formula. For uniqueness, observe

$$\omega^a P^b Q^c e_j = \omega^{a+cj} e_{j+b}.$$

The image of e_0 determines b and a; the image of e_1 then determines c, since ω is a primitive Nth root of unity. □

Proposition 6.32 (The finite central extension)**.** *The centre of* Heis_N *is* $\{\omega^a \mathbb{1} : a \in \mathbb{Z}_N\} \cong \mathbb{Z}_N$*, and there is a nonsplit central extension*

$$1 \longrightarrow \mathbb{Z}_N \xrightarrow{a \mapsto \omega^a \mathbb{1}} \mathrm{Heis}_N \xrightarrow{\pi} \mathbb{Z}_N \times \mathbb{Z}_N \longrightarrow 1, \tag{6.14}$$

Here $\pi(\omega^a P^b Q^c) = (b, c)$ *records the powers of* P *and* Q *modulo* N*. The kernel is the scalar phase subgroup* $\{\omega^a \mathbb{1} : a \in \mathbb{Z}_N\}$*. Thus* Heis_N *is a central extension of* $\mathbb{Z}_N \times \mathbb{Z}_N$ *by the phase group* $\mathbb{Z}_N$*.*

Proof. An element $\omega^a P^b Q^c$ commutes with P exactly when $\omega^c = 1$, and with Q exactly when $\omega^b = 1$. Thus a central element has $b = c = 0$; all remaining scalar matrices are indeed central. Formula (6.13) shows that π is a surjective homomorphism with this centre as kernel, proving exactness.

As in Proposition 6.29, a splitting would require commuting lifts of $(1, 0)$ and $(0, 1)$. Every such pair of lifts has the form $\omega^r P$ and $\omega^s Q$. Multiplying by these scalar phases does not change the relation $QP = \omega PQ$, so they cannot commute. No splitting exists. □

Example 6.33 (The two-site case). For $N = 2$, $\omega = -1$, and $P = X, Q = Z$ are the Pauli matrices of §2.7. Thus

$$\mathrm{Heis}_2 = \{\pm\mathbb{1}, \pm X, \pm Z, \pm XZ\}.$$

This is an eight-element subgroup of the sixteen-element Pauli group P_1 used there; adjoining $i\mathbb{1}$ gives P_1. In particular, $XZ = -ZX$ and $(XZ)^2 = -\mathbb{1}$, even though $X^2 = Z^2 = \mathbb{1}$. The orders of the generators do not determine the orders of their products.

Example 6.34 (The discrete Fourier basis). For $k \in \mathbb{Z}_N$, define the discrete plane wave

$$\psi^{(k)}(z) = \frac{1}{\sqrt{N}} z^k, \qquad \psi^{(k)}(\omega^j) = \frac{1}{\sqrt{N}} \omega^{kj}.$$

At site x_j, the phase $\omega^{kj} = e^{ip_k x_j/\hbar}$ corresponds to momentum $p_k = 2\pi\hbar k/L$. These N wavefunctions form an orthonormal basis, since the geometric series gives

$$\langle \psi^{(k)}, \psi^{(\ell)} \rangle = \frac{1}{N} \sum_{j=0}^{N-1} \omega^{(\ell-k)j} = \delta_{k\ell}.$$

The induced translation and multiplication formulas give

$$(P\psi^{(k)})(z) = \frac{1}{\sqrt{N}}(\omega^{-1}z)^k = \omega^{-k}\psi^{(k)}(z),$$
$$(Q\psi^{(k)})(z) = \frac{1}{\sqrt{N}} z^{k+1} = \psi^{(k+1)}(z).$$

Thus P is diagonal in the wave basis, while Q shifts the wave label. Expanding a state in this basis gives the discrete Fourier pair,

$$\phi(k) = \langle \psi^{(k)}, \psi \rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \psi(\omega^j)\omega^{-kj},$$

$$\psi(\omega^j) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \phi(k)\omega^{kj}.$$

The coefficients $\phi(k)$ are the momentum-space wavefunction, with momentum label k defined modulo N. The operators act on it by

$$(P\phi)(k) = \omega^{-k}\phi(k), \qquad (Q\phi)(k) = \phi(k-1).$$

Since Q moves the basis wave with label $k-1$ to the one with label k, the new coefficient at k is the old coefficient at $k-1$. These are the finite counterparts of the momentum-space formulas for T_a and M_b. The quotient $\mathbb{Z}_N \times \mathbb{Z}_N$ therefore records discrete position and momentum displacements.

These operators act on states. They are symmetries of a Hamiltonian when they commute with it.

The finite construction preserves the exponentiated displacement relation. The canonical commutator itself requires infinite-dimensional operators, since the trace of any finite matrix commutator is zero, whereas $\mathrm{tr}(i\hbar\mathbb{1}) = i\hbar N$.

As in the continuous case, the quotient displacements commute, but their lifts cannot all be chosen to commute. Thus this finite central extension also does not split.

6.5.3. Multiplication in central extensions

Reading. [Moore, §15.2, pp. 431–434].

In Theorem 6.23, a homomorphic section gave the semidirect-product law (6.3). Neither Weyl–Heisenberg extension splits, so we now describe multiplication using a section that need not be a homomorphism.

Consider a central extension

$$1 \longrightarrow A \longrightarrow E \xrightarrow{\pi} G \longrightarrow 1,$$

and identify A with its image in $Z(E)$. Choose a section $s : G \to E$ with $s(e) = e$, so $\pi(s(g)) = g$. From the factorization argument in Theorem 6.23, which uses only $\pi \circ s = \mathrm{id}_G$, we know that every $x \in E$ still has a unique expression

$$x = a\,s(g), \qquad g = \pi(x), \qquad a = x\,s(g)^{-1} \in A.$$

With the additional condition $A \subseteq Z(E)$, the conjugation term in (6.3) becomes $s(g)bs(g)^{-1} = b$. The same multiplication calculation therefore simplifies to

$$\big(a\,s(g)\big)\big(b\,s(h)\big) = ab\,s(g)s(h).$$

The remaining step in the split case was to replace $s(g)s(h)$ by $s(gh)$. Our section need not preserve multiplication. Instead, since

$$\pi\big(s(g)s(h)\big) = gh = \pi\big(s(gh)\big),$$

the elements $s(g)s(h)$ and $s(gh)$ lie in the same kernel coset. Their discrepancy is a unique element of A,

$$s(g)s(h) = c(g,h)s(gh), \qquad c(g,h) = s(g)s(h)s(gh)^{-1} \in A. \tag{6.15}$$

Consequently, writing $a\,s(g)$ as (a,g) gives

$$(a,g)(b,h) = \big(ab\,c(g,h), gh\big).$$

The extension splits precisely when *some* choice of section makes $c(g,h) = e$ for every pair g, h. With that choice the product becomes $(a,g)(b,h) = (ab, gh)$, recovering $E \cong A \times G$.

For the continuous and finite examples, choose $s(a,b) = T_aM_b$ and $s_N(u,v) = P^uQ^v$, respectively. Their multiplication laws give

$$\begin{aligned} c\big((a,b),(a',b')\big) &= e^{\mathrm{i}ba'/\hbar}\mathbb{1} \quad \text{in } \mathscr{W}, \\ c\big((u,v),(u',v')\big) &= \omega^{vu'}\mathbb{1} \qquad \text{in } \mathrm{Heis}_N\,. \end{aligned}$$

These are the phase factors already present in the two Weyl–Heisenberg multiplication laws.

Example 6.35 (The $SU(2)$ extension). Recall the central extension

$$1 \longrightarrow \{\pm\mathbb{1}\} \longrightarrow SU(2) \xrightarrow{R} SO(3) \longrightarrow 1.$$

The kernel $\{\pm\mathbb{1}\}$ is central because scalar matrices commute with every matrix. Let $r \in SO(3)$ be a rotation through π about the third axis, so $r^2 = \mathbb{1}_3$. A splitting would be a homomorphism $s : SO(3) \to SU(2)$ satisfying $R \circ s = \mathrm{id}_{SO(3)}$. The homomorphism condition would require

$$s(r)^2 = s(r^2) = s(\mathbb{1}_3) = \mathbb{1}_2.$$

But $s(r)$ must be a lift of r, and the only two lifts are $\pm\mathrm{i}\sigma^3$. Since $(\sigma^3)^2 = \mathbb{1}_2$, both satisfy

$$(\pm\mathrm{i}\sigma^3)^2 = -\mathbb{1}_2.$$

Neither lift satisfies the required relation. Thus no homomorphic splitting exists, and this central extension does not split. In terms of the factor above, every section with $s(e) = e$ has

$$c(r,r) = s(r)^2 s(r^2)^{-1} = -\mathbb{1}_2.$$

When the lifts $s(g)$ are linear operators and the factors $c(g,h)$ are scalar phases, this is the multiplication law of a *projective representation* of G. In §7.2.1, we will derive the constraints on $c(g,h)$ from associativity and study how these factors change when the operators are rephased.

Quiz 6.5. For $x = (a,b,c)$ and $y = (a',b',c')$ in Heis_N, use (6.13) to find x^{-1} and show

$$xyx^{-1}y^{-1} = \omega^{cb'-c'b}\mathbb{1}.$$

Explain why the quotient is Abelian although the group is not.

Key points

- Normality is exactly what makes coset multiplication well defined. Normal subgroups are precisely kernels of homomorphisms.
- $G/\ker\varphi \cong \operatorname{im}\varphi$: a homomorphism identifies exactly the elements in each kernel coset.
- An extension records a kernel and a quotient together with their maps. These two groups alone do not determine the middle group.
- A splitting gives a semidirect product. Spatial symmetries illustrate both splitting and its failure.
- Position and momentum displacements form a nonsplit central extension of $(\mathbb{R}^2, +)$ by $U(1)$. Clock and shift matrices give the finite counterpart, with quotient $\mathbb{Z}_N \times \mathbb{Z}_N$. A choice of lifts preserves multiplication exactly when all factors $c(g, h)$ are the identity.

Key terms. normal subgroup 正规子群 · quotient group 商群 · simple group 单群 · exact sequence 正合列 · group extension 群扩张 · split extension 分裂扩张 · semidirect product 半直积 · central extension 中心扩张 · Heisenberg group 海森堡群

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